upay Smart Cash-Flow Copilot

A day-20 warning for month-end wallet shortfalls, with reasons and optional suggestions

Project report for AI Hackathon 2026 (DIU CPC × upay)
Track 03: Customer Innovation & Financial Independence

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Repository (public): github.com/mahmudulhasanprotik47/upay-cashflow-copilot

Tested on Python 3.14.8 with pinned requirements. A fresh-clone install test passed on 3 October 2026.

All data in this report is simulated. No upay production data and no customer data were used.

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Where to find each judging criterion

Judging criterion (weight)	Where to look
Problem relevance (20%)	Section 3
AI/ML depth (20%)	Sections 5 and 7
Business/customer impact (20%)	Section 8
Prototype quality (15%)	Section 4; Appendices A and B
Innovation (10%)	Sections 4.5 and 7.4
Scalability and integration (10%)	Section 10
Responsible AI and security (5%)	Section 9

2. Executive summary

Abstract. The Track 03 guideline gives an example customer question: "Why do I always run short before month-end?" The upay Smart Cash-Flow Copilot is a working prototype that tries to answer it before the month ends. On day 20 of a 30-day month it predicts whether a simulated wallet balance will drop below 500 BDT on any day from 21 to 30. It shows a risk band, up to three reasons in English or Bangla, and optional suggestions: a buffer, cash-out fees already paid, and a savings plan when the month looks safe. Nothing moves money; the customer decides. All data is simulated: 1,000 users over six months. On simulated test months, the XGBoost model has a small lead over a simple day-20 balance rule, and the lead is not statistically clear. Its measurable extras are a lowest-balance estimate and a reason for every alert. Warning time is short. Whether alerts change behaviour is untested.

Three headline points

1. **What the customer gets.** A risk band in words, up to three reasons in their own numbers, and suggestions they can take or ignore. In the simulation the alert reached 303 of 464 test-month shortfalls, but 205 of those users were already below 500 BDT on day 20. Among the rest, the model caught 98 of 259, and about 6 in 10 alerts were false alarms.
2. **Why it is worth testing.** Cash-out is large and carries a fee: 23.51% of MFS transaction value in October 2025, and 1.4% on a upay agent cash-out. Our hypothesis is that a customer who sees a tight month coming can keep a buffer and avoid a fee. We have no sourced figure for how often month-end shortfalls happen.
3. **What comes next.** A staged pilot, proposed for discussion with upay: offline replay on governed, anonymised history, then shadow mode, then a randomised test with opted-in customers. The first experiment is to try alert days earlier than day 20, because the median warning time in the simulation was 1 day.

3. User, problem, why now

3.1 Problem statement

> *For a wallet customer whose income arrives late or unevenly, bunched bills and transfers in the first 20 days cause a balance below 500 BDT on at least one of days 21-30. We built a day-20 cash-flow copilot that uses the customer's own transactions from days 1-20 to warn of a likely shortfall, give reasons and suggest a buffer, a fee check or a savings plan, with success measured by the share of user-months that fall below 500 BDT on days 21-30.*

This follows the guideline's template and describes our simulator. The persona, the floor and the causes are assumptions (section 6), not a measured customer segment. The baseline today is the balance itself, which a customer can already see.

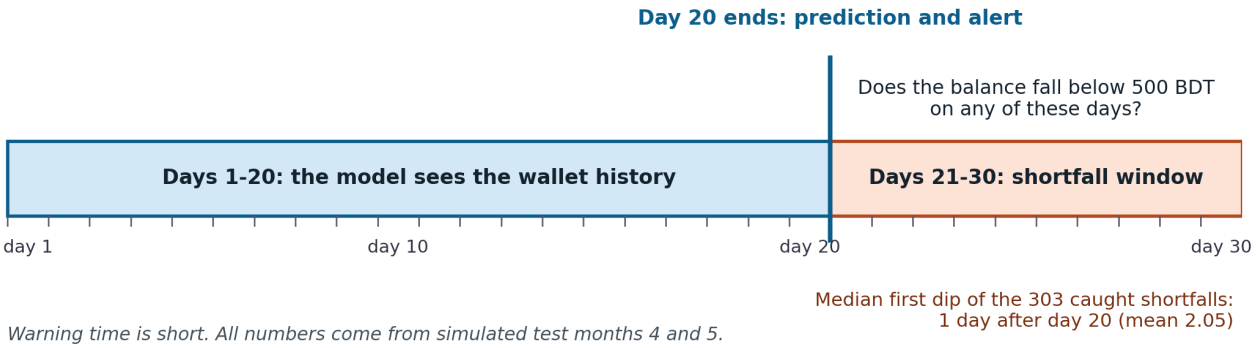


Figure 1. *What happens on day 20. The model sees days 1-20 and gives one alert; the shortfall window is days 21-30.*

3.2 Evidence: cash-out is large and carries a fee

Table A: Sourced context on cash-out (web sources, accessed 3 October 2026; links in Appendix D)

Fact	Source
October 2025: cash-out was 23.51% of MFS transaction value; cash-in was 26.10%.	Bangladesh Bank data via The Financial Express
Same month: 174.56 million cash-out transactions worth Tk 372.23 billion, across providers including bKash, Rocket and upay.	Same article
In 2024, digital payments were 47% to 56% of payment volume, lowest in December (47%, amid year-end cash withdrawals). Non-digital methods were 60% to 80% of payment value.	Bangladesh Payment Systems Report 2024, via The Daily Star
upay agent cash-out, "Primary" type (the only type tested): 7, 14, 70 and 280 BDT on 500, 1,000, 5,000 and 20,000 BDT. That is 1.4%.	upay's official charge calculator (Screenshot 15)
An earlier news report: 1.4% from agent points, including VAT and taxes.	The Business Standard (tbsnews.net)

These facts show that cash-out is large and costs the customer money. They do not show how common month-end shortfalls are. We found no sourced shortfall rate, and the 23.3% in our simulator is a tuned setting, not evidence.

3.3 Why now

A wallet ledger already holds the inputs for a forecast: balance, money in, money out, bills and cash-outs. XGBoost and SHAP, both named in the guideline's suggested methods, can turn them into a warning with reasons.

upay's App Store description lists send money, add money, cash-out at agents and ATMs, utility bills, recharge and store payments; release notes mirrored on APKMirror add bus, launch and air ticket booking. We found no savings, budget or cash-flow feature in those public descriptions. bKash offers DPS savings in its app (Appendix D, W6).

3.4 Nine-step logic chain

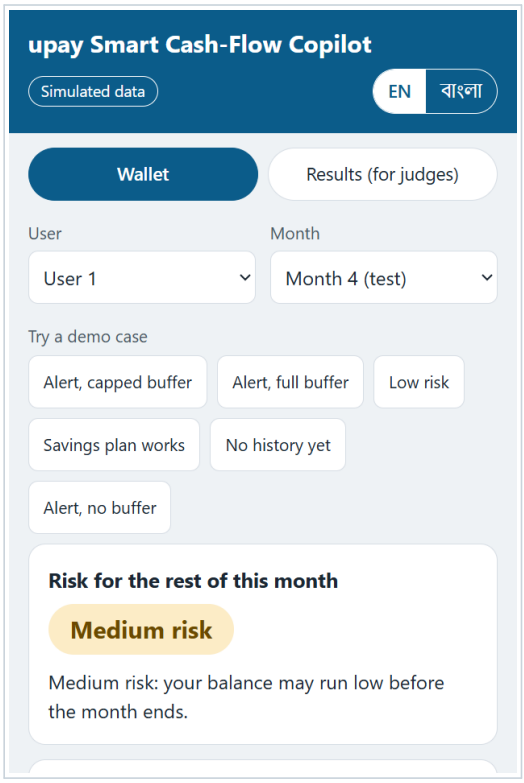
Table B: The guideline's nine steps and our answers

Step	Guideline expects	Our answer	Section
1 User	Persona	Wallet customer with salaried, gig or small-business income. An assumption in the simulator.	3.1, 6
2 Problem	Statement and baseline	Balance below 500 BDT on any of days 21-30. Baselines: a day-20 balance rule and a last-month rule.	3.1, 7
3 Why now	Data/AI opportunity	The ledger holds the inputs. Cash-out is large and carries a fee.	3.2, 3.3
4 Solution	Product flow, prototype	Risk band, reasons, three suggestions, month summary, what-if. Working API and web page.	4
5 AI role	The model's task	Prediction (XGBoost, plus a helper model) and explanation (SHAP). Suggestions are rules.	5
6 Impact	Metric and target	Share of user-months below the floor on days 21-30; cash-out fees per user-month. Targets come from a baseline on governed data. None is invented here.	8
7 Data	Synthetic data strategy	Seeded simulator, 1,000 users, 6 months, assumptions documented.	6
8 Validation	Offline metric and experiment	Time split, two baselines, bootstrap intervals, leakage, direction and fairness checks. A staged pilot proposal.	7, 8
9 Scale	Integration and governance plan	Stateless API, one JSON contract. Authentication, audit logs and monitoring are planned, not built.	9, 10

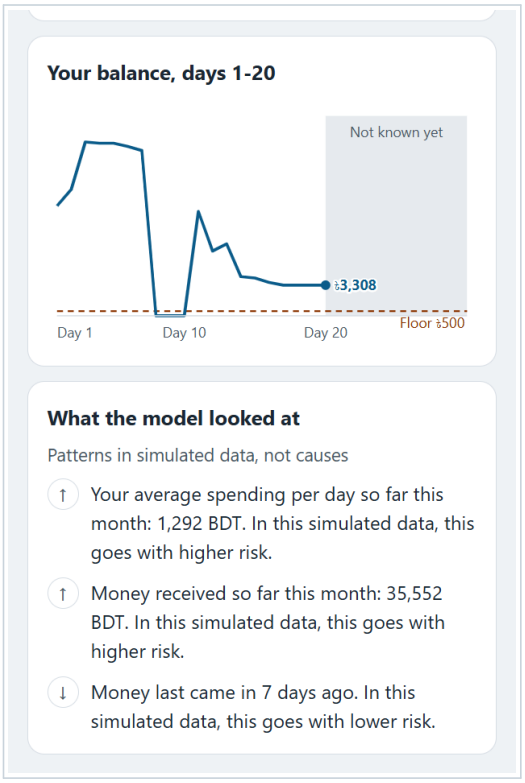
4. Solution and product flow

The prototype is one web page with two tabs, served by a FastAPI service (Appendix B). The Wallet tab is the customer view. The Results tab is for judges. Six one-click demo cases load prepared examples (Appendix A).

4.1 Wallet tab: the day-20 view



Screenshot 01. *Wallet tab, user 1, month 4: "Simulated data" badge, EN/Bangla toggle, six demo-case buttons, Medium risk card.*



Screenshot 02. *Balance chart for days 1-20 with the 500 BDT floor and the "Not known yet" area; day-20 balance 3,308 BDT; three reasons with arrows.*

The first card gives the risk as a band: low, medium or high. The app never shows a probability, because the model is over-confident above 0.2 (section 7).

The chart shows days 1-20 against the 500 BDT floor, with days 21-30 greyed out as "Not known yet". Below it are up to three reasons, each with the customer's own number and a direction arrow. In demo case 1 they are: spending of 1,292 BDT a day (higher risk), 35,552 BDT received (higher risk), and money last received 7 days ago (lower risk). The card says these are patterns in simulated data, not causes.

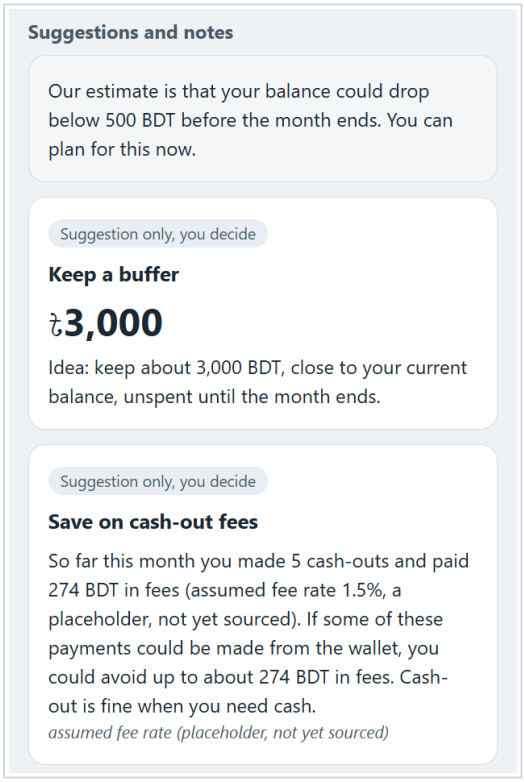
With an alert, the app also shows the estimated lowest balance for days 21-30 with its typical error, but only when the estimate is below 500 BDT (Screenshot 05).

4.2 Three suggestions, all rules

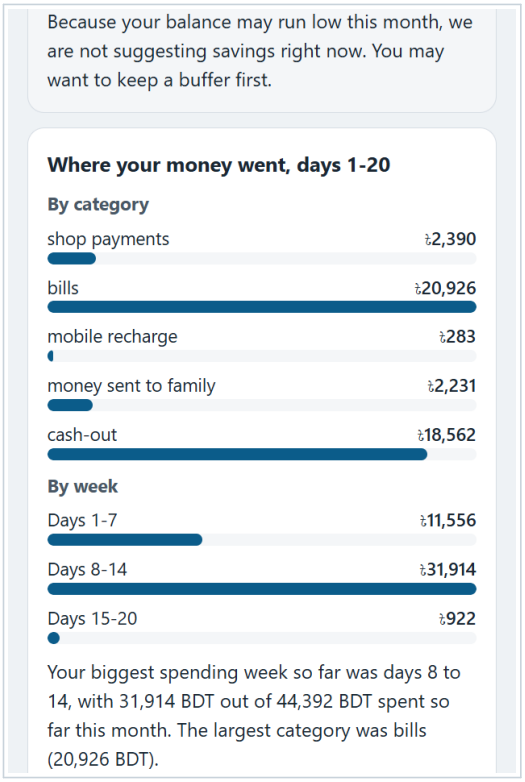
After the prediction comes an action the customer can take. Each suggestion is tagged "Suggestion only, you decide", and nothing happens automatically. All three are fixed rules in `src/rules/suggestions.py`, not model outputs.

Table C: The three suggestion rules, with demo case 1 (user 1, month 4, simulated)

Suggestion	Rule	Demo case 1
Buffer	Only with an alert. 3 days of the user's average daily spending, rounded up to 500 BDT, at least 500, never above the day-20 balance rounded down to 500. If that cap is 0, no buffer.	3 × 1,292 = 3,876, rounded up to 4,000. The 3,308 balance caps it at 3,000 BDT.
Cash-out fee saver	Shown with at least 2 cash-outs on days 2-20. States the count and the fees already paid.	5 cash-outs, 274 BDT at the 1.5% placeholder rate.
Savings plan	Goal of 1-500,000 BDT over 1-24 months. Capacity is half of the user's average monthly net over earlier months. Never offered with an alert.	Blocked by the alert.



Screenshot 03. *Shortfall note, buffer card (3,000 BDT, "Suggestion only, you decide"), cash-out saver card (5 cash-outs, 274 BDT in fees, placeholder label).*

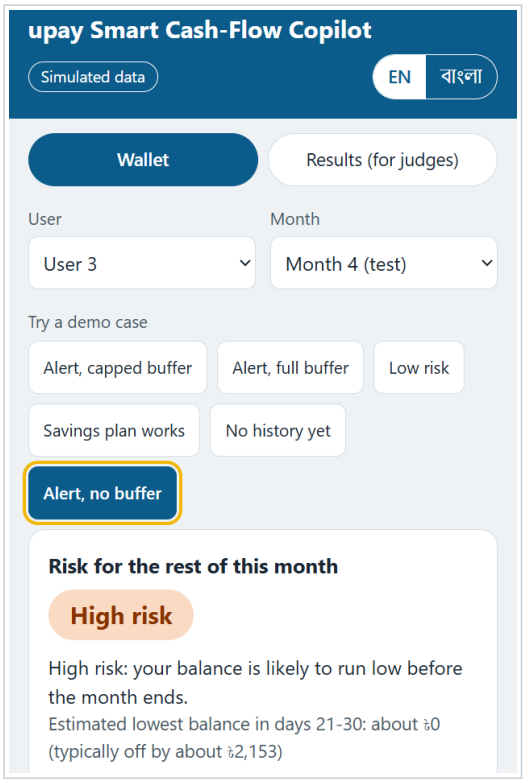


Screenshot 04. *Savings-blocked note and "Where your money went, days 1-20", with bars by category and by week.*

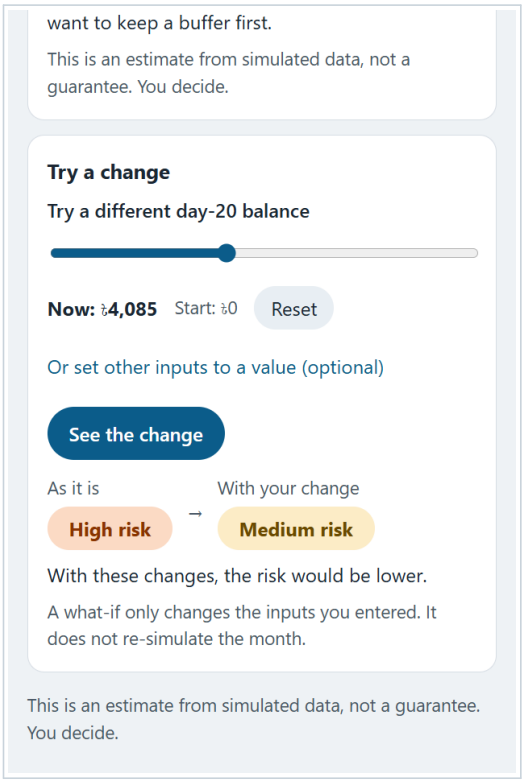
Average daily spending covers shop payments, bills, recharge and family sends on days 1-20, divided by 20. It excludes cash-outs. *Total spent* in the month summary includes cash-outs and their fees. For demo case 1: 44,392 minus 18,562, divided by 20, gives 1,292. The summary also names the biggest week (days 8-14, with 31,914 BDT) and the largest category (bills, 20,926 BDT).

The buffer has a hard limit. In a self-check sample of 300 simulated alerts, 164 got a buffer (minimum 500, median 1,000, maximum 9,500 BDT). The other 136 got none, because the day-20 balance was already under 500 BDT.

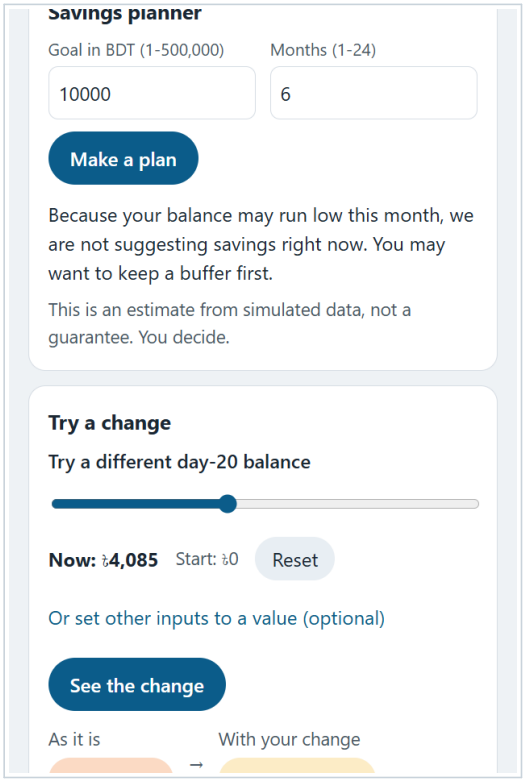
4.3 What-if, and Bangla



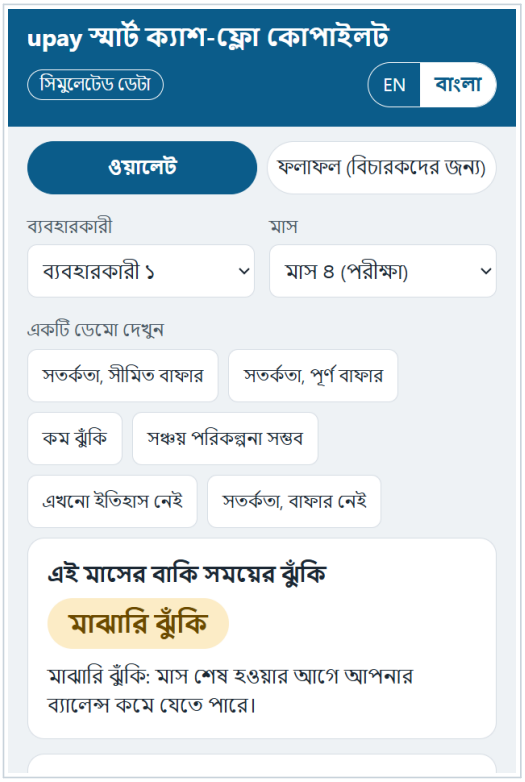
Screenshot 05. *Demo case 6 (user 3, month 4): High risk; estimated lowest balance about 0 BDT, typically off by about 2,153 BDT.*



Screenshot 07. *What-if result: High risk as it is, Medium risk with the change, and the "does not re-simulate the month" note.*



Screenshot 06. *Savings planner (10,000 BDT, 6 months) answered with the "not suggesting savings right now" message; what-if slider (now 4,085, start 0).*

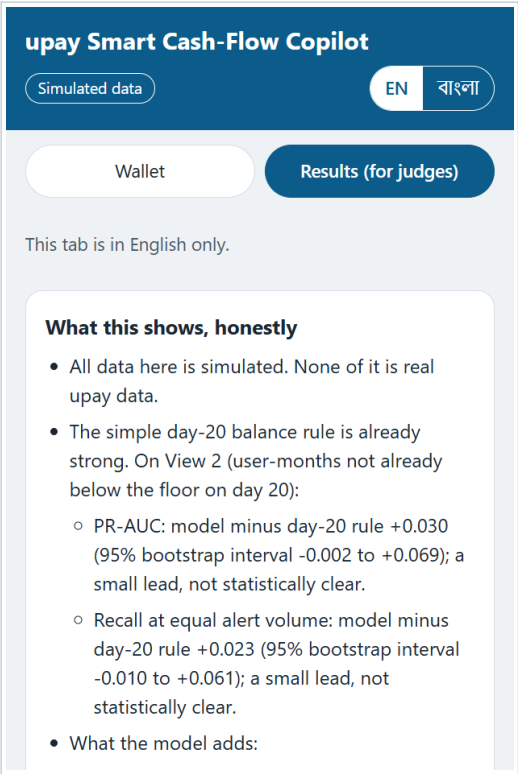


Screenshot 08. *The Wallet tab in Bangla.*

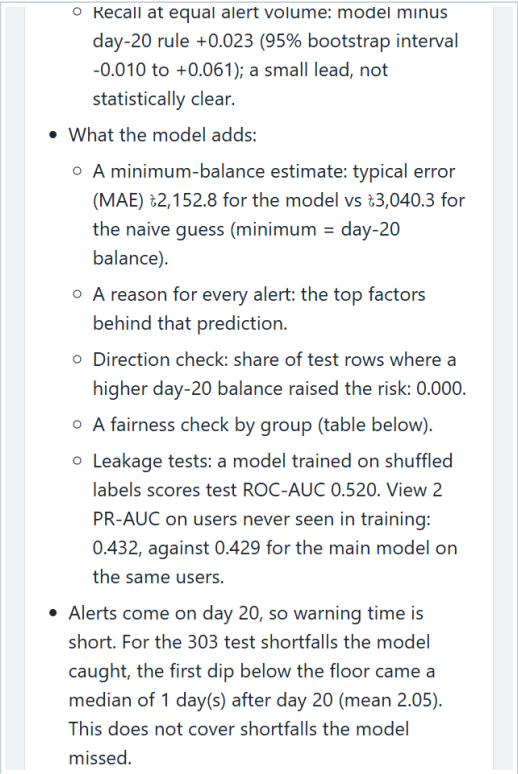
The what-if slider lets the customer try a different day-20 balance and compare the band before and after. In demo case 6, moving it from 0 to 4,085 BDT changes High to Medium. A what-if changes only the inputs entered and does not re-simulate the month.

The EN/Bangla toggle switches the whole Wallet tab. Every message is a fixed template in both languages.

4.4 Results tab



Screenshot 09. *Results tab: the English-only note and the "What this shows, honestly" summary with both intervals.*



Screenshot 10. *Results tab: lowest-balance error, reasons, direction check, fairness check, leakage tests and the lead-time statement.*

The Results tab is in English only, on purpose. It opens by saying that all data is simulated and that the simple day-20 rule is already strong. It then shows both intervals, the model's extras, the SHAP chart, the fairness table and the impact card. Sections 7 and 8 report the same numbers.

4.5 Fit with Track 03, and what is different

The guideline's Track 03 table has four example customer questions: saving toward a goal, running short before month-end, reducing cash-outs, and not understanding transactions. The savings planner, the risk band with reasons, the cash-out saver and the month summary each answer one. Each answer is partial. For example, the saver shows fees paid but does not say which cash-outs a wallet payment could replace (section 11).

Four choices shape the product. It shows a band instead of a probability it cannot stand behind. It gives reasons in the customer's own numbers. It withholds savings suggestions during an alert. Its Results tab opens with the model's weak points. Section 7.4 ties each choice to a check.

5. AI role and architecture

5.1 Architecture

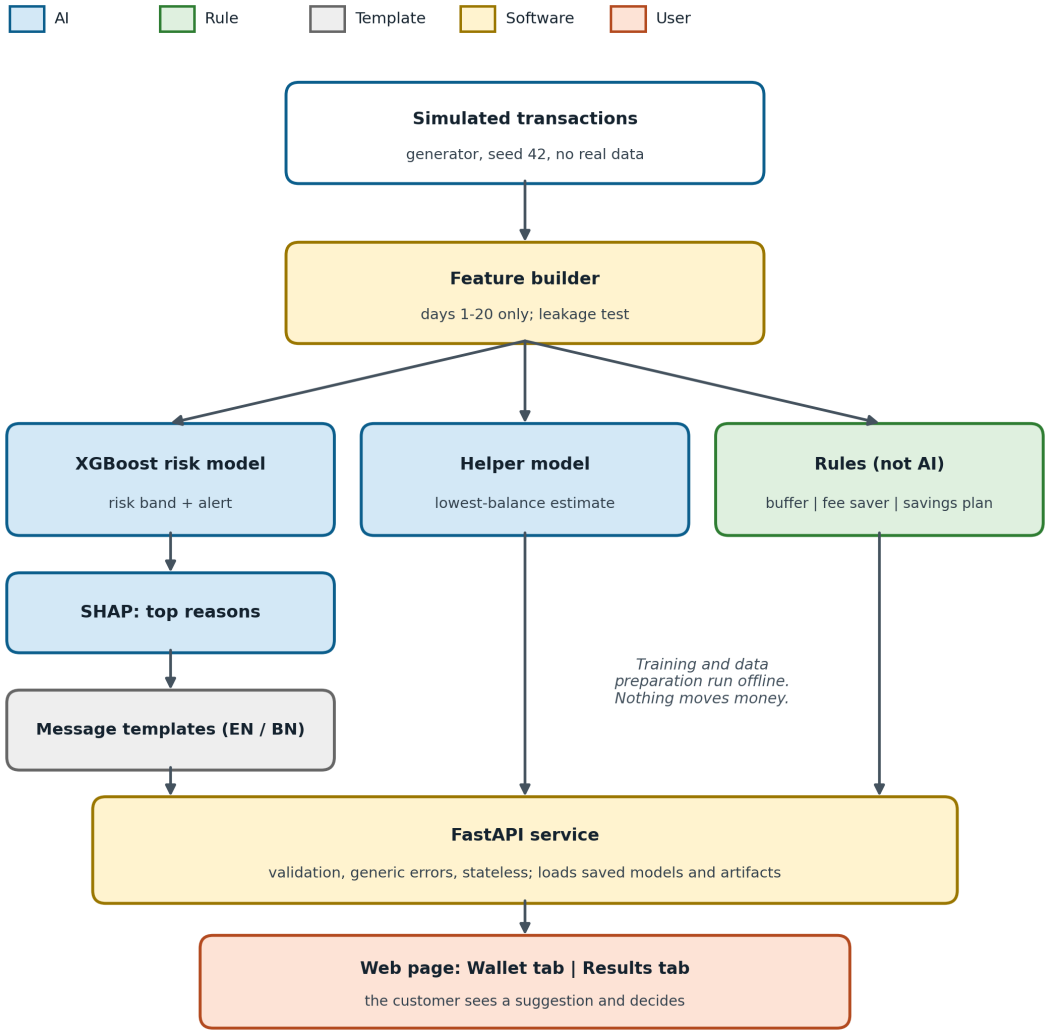


Figure 2. *Architecture, from simulated transactions to a suggestion the user decides on. AI parts are blue, rules green, fixed text grey.*

5.2 What is AI, what is a rule, what is a template

Table D: Every output and what produces it

Output	Type	How it is produced
Risk band and alert	AI prediction	XGBoost classifier: 300 trees, depth 4, learning rate 0.05. Alert when the score is at or above 0.2775.
Lowest-balance estimate	AI prediction	Helper model. Shown only with an alert and only when below 500 BDT, with "typically off by about 2,153 BDT".
Reasons, up to 3	AI explanation	SHAP (TreeExplainer), shown through fixed sentences.
What-if result	AI prediction	The same model, re-scored with the changed inputs.
Buffer, fee saver, savings plan	Rule	Table C.
Month summary	Arithmetic	Sums by category and by week.
All wording, English and Bangla	Template	Fixed templates. No language model anywhere in the product.

The model reads 11 inputs from days 1-20: day-20 balance, lowest balance, average daily spending, money received, total outflow, cash-out count and amount, days since money last came in, number of bill payments, last month's shortfall flag, and money received so far against all of last month. Income band, region and age band are never inputs.

5.3 One model change we disclose

The first trained model had a flaw. For 15.2% of simulated test rows, adding 1,000 BDT to the day-20 balance raised the predicted risk. We added one monotone constraint: a higher day-20 balance can never raise risk. With it, adding 1,000 BDT lowers risk for 62.6% of simulated test rows, raises it for 0.0% and leaves 37.4% unchanged.

We found the flaw in test-month behaviour, so this change was made after seeing the test set. It was made once, on domain grounds, with no further tuning. The alert threshold (0.2775) and the rule cut-off (2,151 BDT) were chosen on training months only, by rolling validation on months 2 and 3.

5.4 The guideline's architecture expectations

Table E: Architecture expectations and our status

Expectation	Status	Evidence and gap
Data preparation separate from model inference	Done	Generation, feature building and training run offline. The API only loads saved models and artifacts.
Business rules distinct from ML predictions	Done	Rules are in `src/rules/`, models in `src/models/`. A self-check confirms the buffer follows its rule.
Traceable and explainable outputs	Partly	Every forecast carries its reasons. Test predictions are saved in `artifacts/`. There is no audit log of alerts shown.
APIs that could connect to a real backend	Partly	One documented JSON contract, input validation, one error shape, stateless. No authentication and no connection to any upay system.
No decision logic inside a free-form LLM prompt	Done	There is no language model. Logic is in code and messages are templates.

6. Data strategy

6.1 What the data is

All data comes from our own generator (`src/data/generator.py`, seed 42). The same seed always gives the same files. It produces 1,000 users over 6 months of 30 days. User ids are counters, and no personal information is involved.

Months 0-3 are training data (4,000 user-months; month 3 is a festival month). Months 4-5 are the test set (2,000 user-months). The split is by time, never random. A shortfall is worked out from the simulated daily balance and is never stored as a label.

6.2 Key assumptions

Every assumption is listed in `docs/synthetic_assumptions.md` (each setting, where it lives in the code, and what is not modelled) and in `docs/data_assumptions.md`. Table F is a summary only. The page header shows a "Simulated data" badge on both tabs.

Table F: Main simulator assumptions

Area	Assumption
Users	Region, age band and income band are drawn from fixed shares. Income type: salaried 55%, gig 30%, small business 15%.
Money in	Salaried users have one payday a month. Gig users get several payments on random days. Business users get roughly weekly receipts.
Money out	Bills, recharge, family sends, shop payments, and agent cash-outs with a fee. On day 1, a balance above one month of income is swept out.
Stress, chance per month	Late salary 12%; thin gig month 20%; bill shock 20%; family emergency 15%. Month 3 is a festival month.
Fee	1.5% on cash-outs only. A placeholder (section 8).
Balance	Never negative. Spending the user cannot afford does not happen.

6.3 What the generator's own checks show

Across all generated data, 23.3% of user-months are shortfalls. The stress settings were tuned to land between 15% and 30%, so this is a calibration aim and not a real-world rate.

The day-20 balance alone separates shortfalls with ROC-AUC 0.851 across all generated data. That is marginally above the generator's own target band of 0.65-0.85, and the check prints a status line for it. On the simulated test months the day-20 rule has ROC-AUC 0.843. A simple balance rule is a strong baseline here, partly because of how we built the simulator.

6.4 What is not modelled, and how this could mislead

Not modelled: loans, fraud, device or location, multiple wallets, real tariff schedules, individual merchants or agents (partial), weekdays or holidays other than the festival month, spending that reacts to a low balance, links between users, and an explicit link to last month's shortfall.

- A pattern in this data may only reflect the simulator's rules.
- Simulated spending never reacts to a low balance or to an alert, so the simulation cannot show behaviour change.
- Spending the user cannot afford is dropped. A shortfall here means a low balance, not a missed bill or a debt.
- Age band has no effect on simulated behaviour, and region only shifts the income-band draw.
- The festival month is in training only. The same 1,000 users appear in the training and test months.
- Every fee amount inherits the 1.5% placeholder.

7. Evaluation

7.1 The story in plain language

All results in this section are simulated. They come from test months 4 and 5, which were not used to train the model or to choose its threshold. Section 5.3 discloses the one change made after looking at test-month behaviour.

We compare the model with two simple rules. The balance rule alerts when the day-20 balance is below 2,151 BDT. The persistence rule alerts when the user had a shortfall last month.

Three things stand out. The balance rule is already strong. The model's lead over it is small and not statistically clear. And 205 of the 464 test-month shortfalls belong to users who were already below 500 BDT on day 20, so they needed no forecast. We therefore report two views. View 2 leaves those users out; it is the stricter test and our headline. View 1 keeps everyone.

7.2 View 2: users not already below the floor

Table 1: View 2, simulated test months. User-months not already below 500 BDT on day 20 (1,783 rows, 259 shortfalls)

Method	PR-AUC	ROC-AUC	Precision	Recall	F1	Recall, equal volume	Precision, equal volume
Model	0.422	0.745	0.430	0.378	0.403	0.324	0.494
Day-20 balance rule	0.392	0.727	0.459	0.301	0.364	0.301	0.459
Persistence rule	0.187	0.601	0.240	0.436	0.310	not available	not available
Model minus rule	**+0.030**					**+0.023**	
95% bootstrap interval	-0.002 to +0.069					-0.010 to +0.061	

Precision, recall and F1 use each method's own threshold. "Equal volume" gives the model and the rule 170 alerts each. The intervals come from a bootstrap over users with 200 resamples.

Both intervals include zero. The model has a small lead, and it is not statistically clear. At its own threshold the model caught 98 of the 259 shortfalls (38%), and about 6 in 10 of its alerts were false alarms. Its precision there is below the rule's (0.430 against 0.459). The persistence rule has the highest recall (0.436), at a precision of 0.240.

7.3 View 1: all user-months, and what it hides

Table 2: View 1, simulated test months. All user-months (2,000 rows, 464 shortfalls)

Method	PR-AUC	ROC-AUC	Precision	Recall	F1	Alerts raised	Shortfalls caught
Model	0.761	0.853	0.681	0.653	0.667	445	303
Day-20 balance rule	0.736	0.843	0.731	0.610	0.665	387	283
Persistence rule	0.302	0.620	0.378	0.478	0.422	not reported	not reported

At equal alert volume (387 alerts each), recall is 0.623 for the model against 0.610 for the rule, and precision is 0.747 against 0.731.

The model's precision is lower than the rule's (0.681 against 0.731). The model also raises more alerts (445 against 387), so part of its recall gain is volume. 29 shortfalls were caught only by the model and 9 only by the rule.

View 1 flatters both methods. Of the 464 shortfalls, 205 (464 minus 259) were users already below 500 BDT on day 20. The model caught all 205, and so did the rule. For those users the alert says what they already know. Take them out and the model's catches fall from 303 to 98 (303 minus 205), out of 259. That is 38%.

7.4 What is different about the product

Four choices set the product apart. Each one rests on a check in this report.

- **A band instead of a probability.** The model is over-confident above 0.2 (Table 3), so the app shows low, medium or high only.
- **Reasons in the customer's own numbers.** Each reason quotes the customer's own value for that input. Self-checks found 0 invented numbers in 1,413 responses and no causal words in 2,768 reason texts.
- **Savings withheld during an alert.** A user with an alert is offered a buffer, never a savings plan. A self-check on 300 simulated rows found no exception.
- **A Results tab that opens with weak points.** Its first card says the simple rule is already strong and shows both intervals (Screenshot 09).

7.5 Lead time, error, calibration and sanity checks

Table 3: Other checks, simulated test months

Check	Result
Lead time for the 303 caught shortfalls: days from day 20 to the first dip below the floor	Median 1 day; mean 2.05 days
Lowest-balance estimate, mean absolute error, all rows	Model 2,152.8 BDT; naive guess 3,040.3 BDT
Lowest-balance estimate, mean absolute error, shortfall rows	Model 2,241.2 BDT; naive guess 3,244.9 BDT
Calibration	Brier score 0.102. In the 0.5-0.6 bin the mean prediction is 0.544 and the actual rate is 0.355 (31 rows)
Leakage test: delete every row after day 20, rebuild the features, compare	Identical. PASS
Model trained on shuffled labels	Test ROC-AUC 0.520
Users never seen in training, View 2 PR-AUC	0.4325, against 0.4292 for the main model on the same users
Direction check: add 1,000 BDT to the day-20 balance	Risk lower for 62.6% of rows, higher for 0.0%, unchanged for 37.4%

The naive guess is that the lowest balance equals the day-20 balance.

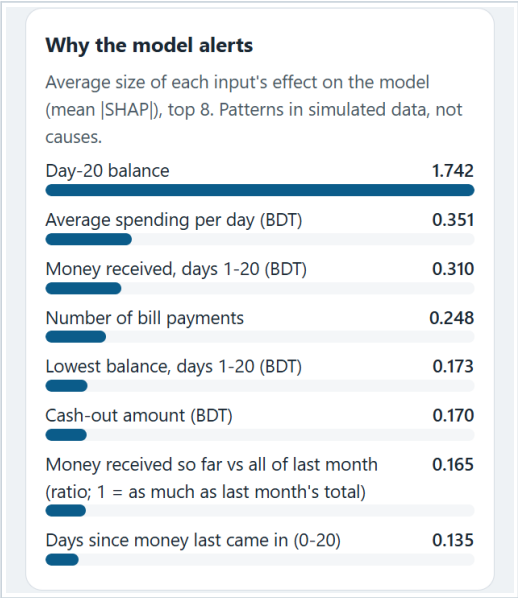
Lead time. Alerts come on day 20, so warning time is short. The median of 1 day covers all 303 caught shortfalls, including the 205 users who were already low. We did not measure lead time for the 98 predictive catches. Testing earlier alert days is the first pilot experiment (section 8.3).

Lowest-balance estimate. The helper model's error is lower than the naive guess, but it is still large next to a 500 BDT floor. The app shows the estimate only with an alert, only when it is below the floor, and always with "typically off by about 2,153 BDT" (Screenshot 05).

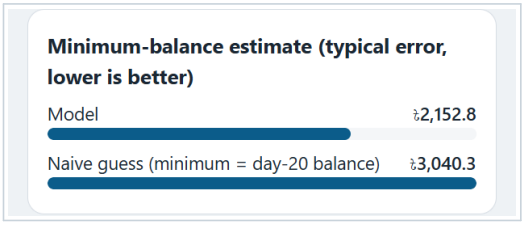
Calibration. The model is over-confident from a predicted risk of 0.2 upward. This is why the app shows bands only.

Sanity and leakage. The features use nothing after day 20. The shuffled-label score is close to chance, as it should be. The unseen-user check trains on 70% of users and tests on the other 30%. The same 1,000 users appear in the training and test months, so this is not a test on a new population. The Results tab lists these two checks under "Leakage tests".

7.6 What drives the predictions



Screenshot 11. *Results tab: top-8 SHAP bars, headed "Patterns in simulated data, not causes".*



Screenshot 12. *Results tab: lowest-balance error bars, model against the naive guess.*

Table 4: Mean absolute SHAP value per input (simulated)

Input	Value	Input	Value
Day-20 balance	1.742	Money received so far against all of last month	0.165
Average spending per day	0.351	Days since money last came in	0.135
Money received, days 1-20	0.310	Total outflow	0.109
Number of bill payments	0.248	Cash-out count	0.059
Lowest balance, days 1-20	0.173	Last month's shortfall	0.018
Cash-out amount	0.170		

"Money received so far against all of last month" is a ratio; 1 means as much as last month's total. The day-20 balance dominates, which fits the strength of the balance rule. These are patterns in simulated data, not causes.

7.7 Fairness check

Table 5: Model alerts by group, simulated test months

Column	Group	Rows	Shortfall rate	Alert rate	Recall	False-alarm rate
Age band	18-25	486	0.247	0.212	0.617	0.079
	26-35	702	0.232	0.208	0.638	0.078
	36-50	572	0.234	0.250	0.664	0.123
	51+	240	0.196	0.221	0.766	0.088
Income band	Low	866	0.244	0.237	0.663	0.099
	Mid	776	0.229	0.209	0.657	0.075
	High	358	0.209	0.218	0.613	0.113
Region	Chattogram	360	0.231	0.231	0.651	0.105
	Dhaka	710	0.215	0.213	0.640	0.095
	Khulna	320	0.231	0.219	0.703	0.073
	Rajshahi	320	0.259	0.237	0.615	0.105
	Sylhet	290	0.245	0.224	0.676	0.078

Table 6: Largest gap between groups (simulated), and how we read it

Column	Recall gap	False-alarm gap	Reading
Age band	0.149	0.045	Likely sampling noise. Age has no effect in the simulator, and the 51+ group has 240 rows and about 47 shortfalls.
Income band	0.050	0.038	Unexplained. Needs a re-check on real data.
Region	0.088	0.032	Unexplained. Needs a re-check on real data.

The model behaves differently across groups. No group has fewer than 150 rows. Age band, income band and region are never model inputs and never appear in user-facing responses; a self-check found none in 1,406 responses. We tuned nothing to close the gaps. This check does not show that the model is fair. It describes one simulated population.

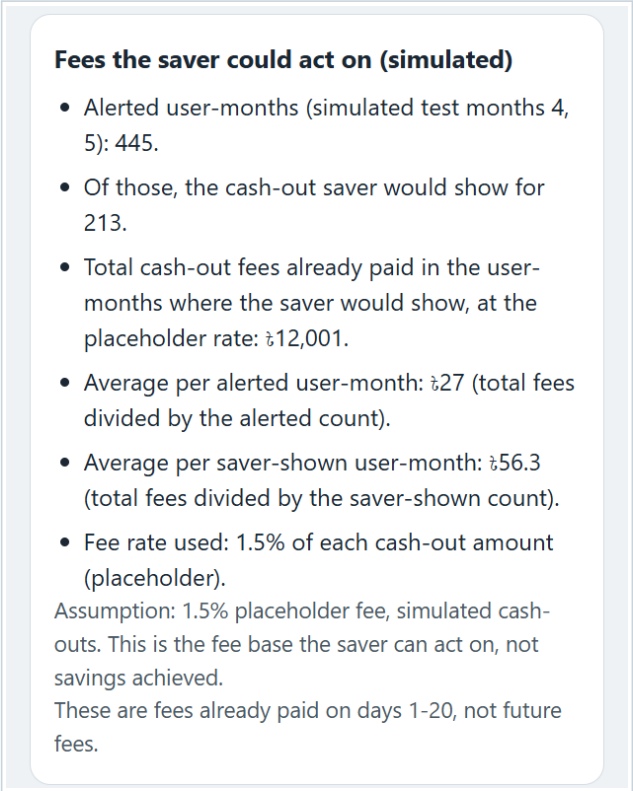
7.8 Does AI add value beyond a simple rule?

Not clearly, on this simulated data. For classification, the model's lead over the day-20 balance rule is small, and both intervals in Table 1 include zero. The measurable extras are the lowest-balance estimate and a reason for every alert. Real data could change this either way.

8. Impact and pilot hypothesis

8.1 What we can report today: a fee base, not savings

The guideline asks for customer and business impact, "not just model performance". The simulation cannot give that. Simulated users never react to an alert, so it cannot show any change in behaviour, balances or fees. What we can report is the fee base that the cash-out saver could act on.



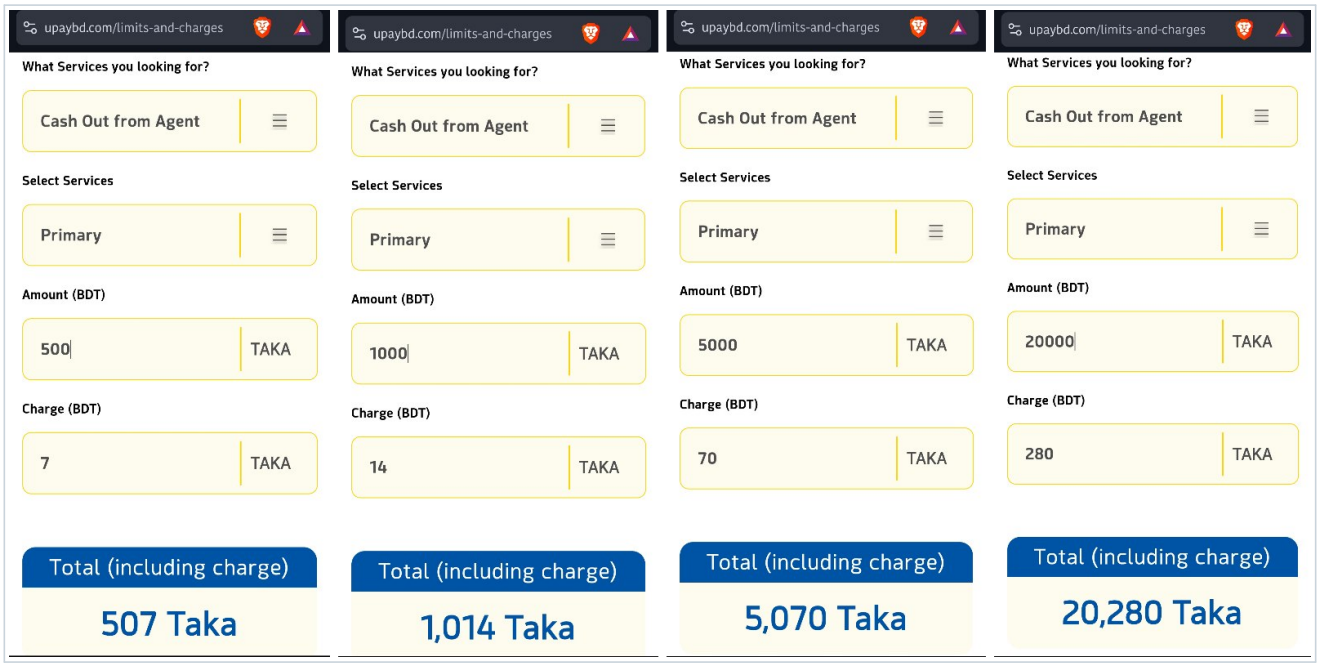
Screenshot 13. *Results tab: impact card "Fees the saver could act on (simulated)", with both assumption sentences.*

Table 7: Cash-out fee base in alerted user-months, simulated test months 4 and 5

Quantity	App output, 1.5% placeholder	Hand estimate at 1.4%, not app output
Alerted user-months	445	445
Of those, user-months where the saver would show	213	213
Cash-out fees already paid in those 213	12,001 BDT	about 11,200 BDT
Average per alerted user-month (total / 445)	27.0 BDT	about 25 BDT
Average per saver-shown user-month (total / 213)	56.3 BDT	about 53 BDT

This is a fee base. These are fees already paid on days 1-20, not future fees and not savings. The 445 alerted user-months include users who were already below the floor on day 20; the artifacts do not report how many.

8.2 The fee rate: a 1.5% placeholder and a published 1.4%



Screenshot 15. *upay's charge calculator at upaybd.com/limits-and-charges: "Cash Out from Agent", type "Primary", for 500, 1,000, 5,000 and 20,000 BDT. Full width.*

Table 8: upay's published charge for "Cash Out from Agent", type "Primary" (official calculator, accessed 3 October 2026)

Amount (BDT)	500	1,000	5,000	20,000
Charge (BDT)	7	14	70	280
Rate	1.4%	1.4%	1.4%	1.4%

The prototype uses 1.5%. We set that placeholder before we found the published figure. upay's calculator shows 1.4% at all four amounts we tested, and we tested only the "Primary" type. An earlier news report also gives 1.4% from agent points, including VAT and taxes.

We did not change the code, because a new rate would change the simulated balances and every reported result. So the app, its on-screen label ("placeholder, not yet sourced") and the impact card still use 1.5%. The right-hand column of Table 7 is a hand estimate: 12,001 × 1.4 / 1.5 gives about 11,200 BDT. Replacing the placeholder and re-running everything is step 3 in section 10.3.

The fee card and the month summary agree with each other. For demo case 1 the summary shows 18,562 BDT of cash-out including fees. 18,562 minus 18,562 / 1.015 is 274 BDT, the fee on the card.

8.3 Pilot proposal

This is a proposal for discussion with upay. Nothing in it is agreed, and we predict no result.

Hypothesis. Alerts plus suggestions reduce the share of users who drop below the floor late in the month.

Table G: Proposed pilot stages

Stage	What happens	Output
0. Offline replay	On governed, anonymised history: re-run the leakage, fairness and direction tests, measure real alert timing, and compare day 20 with earlier alert days.	A real baseline and a real fee schedule
1. Shadow mode	The model scores users, but nobody sees alerts.	Predicted outcomes compared with actual ones
2. Controlled test	Opted-in users, with consent, are randomly assigned to treatment (alerts and suggestions) or control (no alerts). It runs for several monthly cycles. We propose at least 3, so each user passes day 20 three times.	The metrics below

- **Primary metric:** share of user-months below the floor on days 21-30.
- **Secondary metrics:** cash-out fees paid per user-month, how often suggestions are opened, and user-reported confidence.
- **Guardrails:** opt-outs, complaints, false-alarm rate by group, and no pressure wording.

- **Targets and sample size:** set from the Stage 0 baseline before the test starts. Sample size comes from a power calculation on that baseline. We continue only if the primary metric improves and the guardrails hold.

9. Responsible AI and security

9.1 The guideline's seven principles

Table H: The seven principles in the guideline (section 14): what we did, and the gap

Principle	What we did	Honest gap
Privacy	Only simulated data from our own generator. User ids are counters.	No consent or data-minimisation design yet. Both come first in any pilot.
Explainability	Every forecast carries up to three reasons in the customer's own numbers, in English and Bangla.	Reasons are patterns, not causes. Some look odd, such as more money received going with higher risk.
Fairness	We compared alert rate, recall and false-alarm rate across age band, income band and region (Tables 5 and 6).	Gaps exist, and two are unexplained. The check does not show that the model is fair.
Security	Input validation, generic errors, CORS limited to localhost, and no language model (Table I).	No authentication, encryption, audit logs or rate limiting.
Human oversight	The customer makes every decision. The alert threshold sits in one setting, and the model card says a person should review any change together with the fairness numbers.	No operator review process is built.
Transparency	Predictions (risk band, lowest-balance estimate), assumptions ("Simulated data" badge, placeholder fee label) and explanations (reasons card) are separate on screen, and each is labelled.	The on-screen fee label still says "not yet sourced" (section 8.2).
No harmful automation	Nothing moves money. The app approves and denies nothing. Every suggestion and every savings plan carries `requires_user_confirmation: true` and `auto_action: false`.	A wrong alert can still worry a customer. In View 2, about 6 in 10 alerts were false alarms.

9.2 Security: implemented and planned

Table I: Security in the prototype

Implemented	Planned, not implemented
Input validation on every endpoint; extra fields rejected; numbers must be JSON numbers	User authentication and per-user authorisation, so a user sees only their own data
Generic errors that never echo input or show stack traces	Encryption in transit and at rest
CORS limited to localhost origins	Audit logs of alerts shown and suggestions viewed
No probability returned	Rate limiting and secrets management
Fairness numbers only in /model-results and /results	Role-based access to the Results and fairness data
API text rendered as text in the page	Monitoring for data drift and fairness drift
Pinned dependencies; no language model, so no prompt-injection surface	Incident response

The 500-error path is untested, and no load test was done. In production the model inputs would come from the ledger, not from users, so the what-if slider would be a user-facing exploration only. If a language model is ever added, it should be grounded in structured evidence, with decision logic kept out of the prompt, as the guideline asks.

9.3 Track 03: no manipulation, no hidden fees

The track asks teams to avoid manipulative recommendations, hidden fees and designs that push spending. Our templates use no urgency or pressure wording, and a self-check scans all 57 message keys in both languages for banned words. Fees are shown as plain BDT amounts, and the assumed fee rate is labelled a placeholder on screen. The saver card says cash-out is fine when the customer needs cash. No suggestion promotes a product or a purchase, and savings are never suggested during an alert.

10. Scale, integration and path to real data

10.1 What exists today

The service is a stateless FastAPI application with one JSON contract (`docs/api_contract.md`): a documented schema, input validation and one error shape. Scoring is one prediction per user per month, at day 20, so it can run as a batch job. Every suggestion and every savings plan carries `requires_user_confirmation: true` and `auto_action: false`.

On a laptop with 1,000 simulated users, every call returned in under 1 second, and start-up took 1.15 s. No load test was done. No connection to any upay system exists.

10.2 Product readiness checklist

Table J: The guideline's product readiness checklist (section 13), with our status

Checklist item	Status	Why
The user problem is frequent or economically meaningful	Not shown	Cash-out is large and carries a fee (Table A). We have no sourced rate for month-end shortfalls.
AI adds value beyond a simple deterministic rule	Not clear	The lead is small and both intervals include zero (section 7.8). The extras are the lowest-balance estimate and the reasons.
A clear action follows the prediction	Yes, in the prototype	Buffer, fee saver and savings plan; the customer decides. Users already under 500 BDT cannot be offered a buffer.
The business benefit can be measured	Defined, not measured	The pilot metrics are set (section 8.3). Today we have a simulated fee base only.
The model can be validated with future real-world data	Planned	Stage 0 re-runs the same tests on governed history.
Privacy, fairness, explainability and security can be addressed	Partly	Reasons and a fairness check exist. Authentication, encryption, audit logs and consent are planned.
The system can integrate into a real digital-service workflow	Untested	Stateless API, one JSON contract, batch scoring at day 20. Nothing is connected to upay.

10.3 Path to real data

1. Run an opted-in pilot under a data-sharing agreement.
2. Re-run the leakage, fairness and direction tests on real history.
3. Replace the placeholder with the sourced fee.
4. Measure real alert timing and test earlier alert days.
5. Put consent, data minimisation and security first.

These five steps are our proposal. upay has agreed to none of them, and the guideline itself calls the follow-on stage "not a promise of access or deployment". In the guideline's pathway, Stage 0 of our pilot corresponds to controlled validation, and Stages 1 and 2 to the POC and pilot assessment.

11. Limitations and what we do not claim

Limitations

- All results are simulated. A pattern may only reflect the simulator's rules.
- We have no sourced rate for month-end shortfalls. The simulated 23.3% is a tuned setting.
- The model's lead over the day-20 balance rule is small and not statistically clear. At its own threshold its precision is lower than the rule's in both views.
- 205 of the 303 caught shortfalls were users already below the floor on day 20. In View 2 the model caught 98 of 259, and about 6 in 10 alerts were false alarms.
- Warning time is short: a median of 1 day. Lead time for the 98 predictive catches was not measured.
- The lowest-balance estimate is typically off by about 2,153 BDT. In a self-check sample of 300 simulated alerts, the estimate was at or above the floor for 161, so no estimate was shown.
- In a self-check sample of 300 simulated alerts, 136 could get no buffer.
- The model is over-confident above 0.2, so the app can show bands only.

- One model change was made after looking at test-month behaviour (section 5.3).
- The festival month is in training only, so the model is untested on one. The same 1,000 users appear in the training and test months.
- The age gap in the fairness check is likely sampling noise. The income-band and region gaps are unexplained and need a re-check on real data.
- The code uses a 1.5% fee placeholder. The published rate is 1.4%, and we tested only the "Primary" type.
- Simulated users never react, so we have no evidence that alerts or suggestions change behaviour.
- The suggestions are partial. The fee saver does not say which cash-outs a wallet payment could replace. The month summary covers days 1-20 of one month and shows category totals only.
- No load test was done, the 500-error path is untested, there is no authentication, and nothing is connected to upay. The code was tested on Python 3.14.8 only.

What we do not claim

- We do not claim a clear win over the simple rule.
- We do not claim that month-end shortfalls are common among upay customers.
- We do not claim any amount saved. The impact card is a fee base.
- We do not claim the model is fair.
- We make no claim about upay's systems, plans or internal features. We looked only at public app descriptions.
- We make no claim about cost or revenue for upay. We have not modelled either.

12. AI tool use

Claude Code (Anthropic) wrote most of the code from specifications written by Mahmudul Hasan, who reviewed every change and ran all tests. Claude (chat) helped with planning and with drafting this report, which the team checked against the test results. Other team members also used Claude (chat) for research and slide preparation. In the product, AI is used only for the XGBoost prediction and the SHAP reasons; there is no language model, and all user messages are fixed templates. The official guideline has no rule on AI-tool use.

Appendix A: Demo cases

Table K: The six one-click demo cases (all simulated)

#	Button	User, month	What it shows
1	Alert, capped buffer	User 1, month 4	Medium risk. Day-20 balance 3,308 BDT, buffer 3,000 BDT. The cap is the balance rounded down to 500 BDT.
2	Alert, full buffer	User 8, month 4	An alert with a buffer of 1,000 BDT that the balance does not cap.
3	Low risk	User 1, month 5	No alert.
4	Savings plan works	User 19, month 4	Low risk. A goal of 10,000 BDT over 6 months is feasible.
5	No history yet	User 1, month 0	The first month. There are no earlier months to learn from.
6	Alert, no buffer	User 3, month 4	High risk. Estimated lowest balance about 0 BDT, typically off by about 2,153 BDT. The balance is too low for a buffer.

Appendix B: Repo map and how to run

```
src/data/generator.py             seeded data generator
src/features/build_features.py    features from days 1-20
src/models/train_liquidity.py     model training
src/models/predict.py             prediction
src/rules/suggestions.py          buffer, fee saver, savings plan (rules)
src/i18n/messages.py              English and Bangla templates
src/api/main.py                   FastAPI app
src/api/service.py                service logic and the 16 self-checks
src/config.py                     settings
frontend/index.html               the web page (Wallet and Results tabs)
artifacts/                        metrics, fairness, SHAP, trained models,
                                  test predictions, threshold
docs/                             model_card, target_definition, api_contract,
                                  data_assumptions, synthetic_assumptions,
                                  messages_preview
requirements.txt, README.md       pinned dependencies, quick start
```

How to run (tested on Python 3.14.8)

```
python -m venv .venv
.venv\Scripts\Activate.ps1              (Windows PowerShell)
python -m pip install -r requirements.txt
python -m src.data.generator
python -m src.api.service          (expects 16 PASS)
python -m uvicorn src.api.main:app --host 127.0.0.1 --port 8000
```

Then open http://127.0.0.1:8000/ in a browser. Use months 4 and 5 for demos.

Endpoints. GET /health; GET /users; GET /users/{id}/months/{m}/forecast; GET /users/{id}/months/{m}/summary; POST /savings-plan; POST /whatif; GET /model-results; GET /results; and the page at /.

Self-checks (16, all pass). They include: templates render with no placeholders or banned words (57 keys in 2 languages); no fairness columns in 1,406 responses; every number in a message appears in the response data (1,413 responses, 0 invented); no causal words in 2,768 reason texts; adding 1,000 BDT in a what-if never raised risk (0 of 200); the buffer follows its rule on 300 simulated alerts; every call under 1 second.

Appendix C: Claim check

Tags: **[R]** shown in the app's Results tab. **[D]** repo document, artifact or self-check output. **[W]** web source (Appendix D). **[H]** hand estimate or our own arithmetic. Every number in a row carries that row's tag.

Table L: Every number in the report and its source

Section	Claim	Value	Tag
2, 3.1, 6.1	Prediction day; month length; floor; target window	Day 20; 30 days; 500 BDT; days 21-30	[D]
2, 6.1	Simulated population	1,000 users; 6 months; seed 42	[D]
6.1	Train and test split	Months 0-3 = 4,000 user-months; months 4-5 = 2,000	[D]
6.2	Simulator settings	Income types 55%, 30%, 15%; stress chances 12%, 20%, 20%, 15%; festival in month 3; day-1 sweep above 1 month of income	[D]
3.2, 6.3, 11	Shortfall share in generated data; tuning aim	23.3%; 15% to 30%	[D]
6.3	Day-20 balance alone; generator's target band	ROC-AUC 0.851; 0.65 to 0.85	[D]
5.2	Model	300 trees; depth 4; learning rate 0.05; 11 inputs	[D]
5.2, 5.3, 7.1	Alert threshold; rule cut-off; validation months	0.2775; 2,151 BDT; months 2 and 3	[D]
5.3, 7.5	Direction check, before and after the constraint	Before: 15.2% higher. After: 62.6% lower, 0.0% higher, 37.4% unchanged	[D]
7.2	View 2 size	1,783 rows; 259 shortfalls	[D]
7.2	View 2, model: PR-AUC, ROC-AUC, precision, recall, F1	0.422; 0.745; 0.430; 0.378; 0.403	[D]
7.2	View 2, balance rule, same order	0.392; 0.727; 0.459; 0.301; 0.364	[D]
7.2	View 2, persistence rule, same order	0.187; 0.601; 0.240; 0.436; 0.310	[D]
7.2	View 2 at equal volume	170 alerts; model recall 0.324, precision 0.494; rule 0.301, 0.459	[D]
7.2	Model minus rule, PR-AUC, with interval	+0.030; -0.002 to +0.069	[R]
7.2	Model minus rule, recall at equal volume, with interval	+0.023; -0.010 to +0.061	[R]
7.2	Bootstrap	200 resamples over users	[D]
7.3	View 1 size	2,000 rows; 464 shortfalls	[D]
7.3	View 1, model, same order	0.761; 0.853; 0.681; 0.653; 0.667	[D]
6.3, 7.3	View 1, balance rule, same order	0.736; 0.843; 0.731; 0.610; 0.665	[D]
7.3	View 1, persistence rule, same order	0.302; 0.620; 0.378; 0.478; 0.422	[D]
7.3	View 1 at equal volume	387 alerts; recall 0.623 and 0.610; precision 0.747 and 0.731	[D]
2, 7.3, 8.1	Alerts raised; shortfalls caught	Alerts 445 and 387; caught 303 and 283; only the model 29; only the rule 9	[D]
2, 7.1, 7.3, 11	Already below the floor on day 20; predictive catches	205 (464 - 259); 98 of 259 (303 - 205)	[D]
2, 7.2, 7.3, 9.1, 11	Share caught in View 2; false alarms	38% (98 / 259); about 6 in 10 (1 - 0.430)	[H]
2, 7.5, 11	Lead time	303 shortfalls; median 1 day; mean 2.05 days	[R]
7.5	Lowest-balance error, all rows	2,152.8 and 3,040.3 BDT	[R]
7.5	Lowest-balance error, shortfall rows	2,241.2 and 3,244.9 BDT	[D]
4.3, 5.2, 7.5, 11	Typical error shown in the app	About 2,153 BDT	[D]
4.1, 7.4, 7.5, 11	Calibration	Brier 0.102; over-confident from 0.2; 0.5-0.6 bin: 0.544 predicted, 0.355 actual, 31 rows	[D]
7.5	Shuffled-label model	ROC-AUC 0.520	[R]
7.5	Unseen users; user split	PR-AUC 0.4325 and 0.4292; 70% and 30%	[D]
7.6	SHAP, top 8 inputs	1.742; 0.351; 0.310; 0.248; 0.173; 0.170; 0.165; 0.135	[R]
7.6	SHAP, other 3 inputs	0.109; 0.059; 0.018	[D]
7.7	Fairness	Every value in Tables 5 and 6; 150-row minimum; 51+ group: 240 rows, about 47 shortfalls	[D]
8.1	Impact card	445; 213; 12,001 BDT; 27.0 BDT; 56.3 BDT; 1.5%	[R]
8.1, 8.2	Fee base at 1.4%	About 11,200 BDT (12,001 × 1.4 / 1.5); about 25 BDT; about 53 BDT	[H]

Section	Claim	Value	Tag
2, 3.2, 8.2	upay agent cash-out charge, "Primary"	7, 14, 70, 280 BDT on 500, 1,000, 5,000, 20,000 BDT; 1.4%	[W]
3.2, 8.2	Earlier news report on upay's agent charge	1.4%	[W]
2, 3.2	MFS statistics, October 2025	23.51%; 26.10%; 174.56 million; Tk 372.23 billion	[W]
3.2	Digital share of payment volume in 2024; non-digital share of payment value	47% to 56%; 60% to 80%	[W]
4.1, 4.2, App. A	Demo case 1	3,308; 1,292; 35,552; 7 days; 3,000; 5 cash-outs; 274; 44,392; 18,562; 20,926; 31,914	[D]
4.2, 8.2	Demo case 1 arithmetic	3 × 1,292 = 3,876, rounded up to 4,000; (44,392 - 18,562) / 20 = 1,292; 18,562 - 18,562 / 1.015 = 274	[H]
App. A	Demo cases 2 and 4	Buffer 1,000 BDT; goal 10,000 BDT over 6 months	[D]
4.3, App. A	Demo case 6, Wallet tab (Screenshots 05 to 07)	Lowest balance about 0 BDT; what-if from 0 to 4,085 BDT: High to Medium	[R]
4.2	Rule settings	3 days of spending; 500 BDT steps; at least 2 cash-outs on days 2-20; goal 1 to 500,000 BDT; 1 to 24 months; half of average monthly net	[D]
4.2, 11	Self-check samples of alerts	300 alerts; 164 with a buffer (minimum 500, median 1,000, maximum 9,500 BDT); 136 with none; 161 with the estimate at or above the floor	[D]
7.4, 7.7, 9.3, App. B	Other self-check counts	16 checks; 57 keys in 2 languages; 1,406 responses; 1,413 responses, 0 invented; 2,768 texts; 0 of 200 what-ifs; 300 rows, no savings on an alert	[D]
10.1, App. B	Speed	Under 1 second per call; start-up 1.15 s	[D]
Cover, 11, App. B	Environment	Python 3.14.8	[D]
8.3	Pilot length (a proposal, not a result)	At least 3 monthly cycles	[H]

The demo case 6 row is app output from the Wallet tab, not the Results tab. Counts taken from the guideline itself (nine steps, seven principles, seven checklist items, four example questions) are not listed.

Appendix D: Evidence log

Table M: Web sources (all accessed 3 October 2026)

#	Source	Link	What it supports
W1	The Financial Express, "MFS transactions maintain rising trend in Oct '25", citing Bangladesh Bank data	thefinancialexpress.com.bd/trade/mfs-transactions-maintain-rising-trend-in-oct-25	October 2025: cash-out was 23.51% and cash-in 26.10% of MFS transaction value; 174.56 million cash-out transactions worth Tk 372.23 billion. The statistics cover providers including bKash, Rocket and upay.
W2	The Daily Star, "Digital transactions rise, but cash-based payments still dominate", reporting the Bangladesh Payment Systems Report 2024	thedailystar.net/business/news/digital-transactions-rise-cash-based-payments-still-dominate-4010706	In 2024, digital payments were 47% to 56% of payment volume, lowest in December (47%, amid year-end cash withdrawals). Non-digital methods were 60% to 80% of payment value.
W3	upay, official limits and charges calculator	upaybd.com/limits-and-charges	"Cash Out from Agent", type "Primary": 7, 14, 70 and 280 BDT on 500, 1,000, 5,000 and 20,000 BDT, which is 1.4% (Screenshot 15).
W4	The Business Standard, "UCB's upay offers lowest cash-out fee"	tbsnews.net/node/234661	upay's cash-out charge is 1.4% from agent points, including VAT and taxes.
W5	Apple App Store listing for the upay app	apps.apple.com/app/id1576179329	upay's App Store description lists send money, add money, cash-out at agents and ATMs, utility bills, recharge and store payments. We found no savings, budget or cash-flow feature in it.
W6	The Financial Express, "bKash App: A Nifty Solution to Manage Personal Finance"	thefinancialexpress.com.bd/trade/bkash-app-a-nifty-solution-to-manage-personal-finance	bKash offers DPS savings in its app.
W7	APKMirror, release notes for the upay app (a third-party mirror of the app's what's-new text)	apkmirror.com/apk/upay/upay-send-money-recharge-payment-utility	Release notes mention bus, launch and air ticket booking. No savings, budget or cash-flow feature appears in them.

None of these sources gives a rate for month-end shortfalls.